



Ax-Series EDC Protocol Manual

Ax150i

Ax350i

Ax550i

DOMINO

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For sales, service and inks please contact:

www.buydomino.com

www.domino-printing.com

Domino UK Limited

Trafalgar Way
Bar Hill
Cambridge CB23 8TU
United Kingdom
Tel: +44 (0)1954 782551
Fax: +44 (0)1954 782874
Email: enquiries@domino-uk.com

Domino North America

1290 Lakeside Drive
Gurnee IL.60031
U.S.A.
Tel: +1 847 244 2501
Fax: +1 847 244 1421

AMENDMENT RECORD

Amendment

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PREFACE PROTOCOL MANUAL

This document, Domino Part No. EPT077984, is intended to provide EDC setup information for the Ax-Series printer range.

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PART 1 INTRODUCTION

WHAT IS EDC?

Note

This feature is not available for Ax130i.

EDC (external data capture) enables pre-defined parts of a label (EDC Elements) to be populated with data sent from an external source. External data is stored in a buffer queue within the printer's memory in advance of printing.

Data can be sent to the printer via RS232 or Ethernet connection.

If RS232 is used (EDC Serial), the socket used for EDC cannot be used for any other communication protocols. It will not be possible to use Codenet or InterComm and EDC via the same serial communication socket.

If Ethernet is used (EDC TCP), all communication protocols can be used via the same Ethernet socket.

PRINTING MODES

EDC can be setup to populate the label with data in 1 of 2 different modes:

- Buffer printing
- Batch printing.

What is Buffer Printing?

When printing in buffer mode, multiple data strings are sent to the printer and stored in a FIFO (First In First Out) buffer in advance of printing. Every label/product will get unique information from a different data string.

Buffer printing is commonly used in the food industry to print the weight and value of an item onto packaging, when the weight and value of each item varies.



After a data string is used to populate EDC elements in a label, the data is deleted from the buffer and the next data string is used for the next label. This will continue until the data buffer becomes empty.

There are many options to choose the printer's behavior when the data buffer becomes empty.

To optimise performance, it's best to send the smallest possible number of data strings to the printer's data buffer at one time.

To print data in this mode, the *Persistence* setting which is set when the RS232 or Ethernet connection is setup will need to be disabled. For information about setting the persistence, see [How to Setup an EDC RS232 Connection on page 10](#) or [How to Setup an EDC Ethernet Connection on page 18](#) (depending on the connection type).

What is Batch Printing?

Every label/product in a production run/batch gets the same information from a single data string.

This mode may be used on a production line where the same label is used for different production runs, but only a small change to the label data is needed at the start of each production run. For example, only the batch number of the production run and best before date may change.



To print data in this mode, the *Persistence* setting which is set when the RS232 or Ethernet connection is setup will need to be enabled. For information about setting the persistence, see [How to Setup an EDC RS232 Connection on page 10](#) or [How to Setup an EDC Ethernet Connection on page 18](#) (depending on the connection type).

DATA STRING FORMAT

Start and End Markers

To identify the start and end of each data string, a start and end marker is used. Any characters can be used as the start and end marker. When setting up the RS232 or Ethernet connection, the markers used by the system need to be input in the printer's settings to let the printer identify them.

Typical start markers are <STX> or <ESC>

Typical end markers are <ETX> or <EOT>

For example:

Start Marker	Data Transmission	End Marker
<STX>	??	<ETX>

Or:

Start Marker	Data Transmission	End Marker
<ESC>	??	<EOT>

Delimiters

If the data string contains more than 1 item of data, separate each item with a delimiter character. Any character can be used as a delimiter character.

A typical delimiter character is the comma symbol: ,

For example:

Start Marker	Index0	Delimiter	Index1	Delimiter	Index2	End Marker
<STX>	123	,	456	,	789	<ETX>

When the EDC element is setup and made in the label creator, input the delimiter character used by the system. This will let the printer identify the delimiter character so that the correct data can be printed.

The specific item of data that will be printed is chosen by setting the Index and Offset settings, when the EDC element is setup and made in the label creator.

For information about setting up and creating an EDC element, see [How to Create and Add an EDC Text Element to a Label on page 30](#).

PART 2 EDC SETUP

HOW TO SETUP AN EDC RS232 CONNECTION

Note If the settings described in this topic are changed when a label containing EDC elements is loaded for printing, re-load the label to print.

For the connection to be successful, it is essential that the settings described in this topic are set correctly and mirror the settings of the device that is used to transmit data to the printer.

To setup and enable the EDC connection via RS232:

1. Select *Home > Setup > Printer network > Protocol settings*.
2. Select the *Protocol Setting* drop down and select *EDC Serial*.
3. Before changing any settings, un-tick (disable) the *Protocol enabled* tick box.

Note The settings cannot be changed, when the protocol is enabled.

4. Select the *Serial Com* drop down setting and select either:

Serial Com Setting	Description
COM1	For communication via the RS232 Pack's RS232 Socket (if fitted).
COM2	For communication via the Extended IO Pack's 37-Way Socket (if fitted).

5. Select *Advanced*.
6. EDC settings described in the table below can be set using the drop down boxes:

Setting Name	Description
Baud rate	Select the baud rate between the printer and the device that the printer is connected to. Range: 1200-115200 Default setting: 9600
Data bits	Select the number of data bits in each character. Range: 7-8 Default setting: 8
Parity	Select the type of parity bit that will be expected. A parity bit is an extra data bit that will be sent with each data character to detect errors in the RS232 transmission. No parity - No parity bit will be expected. Odd - An odd parity bit will be expected. Even - An even parity bit will be expected. Mark - The parity bit will always be 1 but will not be used. Space - The parity bit will always be 0 but will not be used. Default setting: No parity

EDC SETUP

Setting Name	Description
Stop bits	Select the number of stop bits that will be sent at the end of each character to enable the detection of the end of the character. Valid settings: 1, 1.5, or 2 Default setting: 1

Setting Name	Description
Flow control	Select the type of handshake signal that will be used to pause and resume the transmission of data. None - Flow control will not be enabled. None, DTR enabled - - The DTR output line will be asserted (+12V) whenever the printer is ready to receive data. - The DTR output line will be de-asserted (-12V) whenever the printer is not ready for data entry. DTR/DSR - - The DTR output line will be asserted (+12V) whenever the printer is ready to receive data. - The DTR output line will be de-asserted (-12V) whenever the printer is not ready for data entry. - Whenever the DSR input line is asserted (+12V), the printer will be enabled for the sending of data. - Whenever the DSR input line is de-asserted (-12V), the printer will suspend data output. RTS/CTS - - The RTS output line will be asserted (+12V) whenever the printer is ready to receive data. - The RTS output line will be de-asserted (-12V) whenever the printer is not ready for data entry. - Whenever the CTS input line is asserted (+12V), the printer will be enabled for the sending of data. - Whenever the CTS input line is de-asserted (-12V), the printer will suspend data output. XOn/XOff - Note When XOn/Xoff is selected, CTS and RTS signals must not be connected. - The printer will send a single ASCII <XOFF> (013) character whenever it requires data entry to be suspended. - The printer will send a single ASCII (011) character whenever it becomes ready to receive data. - Whenever the printer receives an <XOFF> character it will suspend its data output. Transmission from the printer will be enabled whenever it receives an character. Default setting: None

7. Select *Save*.

8. The settings described in the table below can now be defined. After these settings have been defined, tick (enable) the *Protocol Enabled* tick box.

Setting Name	Description
Encoding	Set the data string's encoding method: • ASCII • UTF-8 • Unicode(BE) • Unicode(LE) Default setting: ASCII
Start marker	Define the start marker that is used to mark the start of printable items in the data string. Default setting: <STX>
End marker	Define the end marker that is used to mark the end of printable items in the data string. Default setting: <ETX>
Packet length	If <i>Data packet type</i> is set to <i>Use specific length</i> this setting defines the length of printable items in the data string that the printer will look for. The length is defined in bytes. Default setting: 1
Data packet type	Set if the start and end of each data string will be marked with start and end markers. Or, set if the printer will look for items of a specific length. Default setting: Use start end marker
Ack type	Setup the acknowledgment that the printer will send after the printer receives a data string: • None - No acknowledgement will be sent. • Default - The letters ACK (41 43 4B) will be sent. • RegExpression - A user defined acknowledgment will be sent (This feature will be enabled in a later software update). • DefaultSpecialChars - ACK (06) will be sent. • BufferDepth - The number of items in the data buffer, or the amount of space in the data buffer will be sent. The behavior is defined by the <i>Ack/Nack buffer depth</i> setting. Default setting: DefaultSpecialChars

Setting Name	Description
Ack/Nack buffer depth	Note This setting is only valid, if <i>Ack type</i> is set to <i>BufferDepth</i>. Select if the number of items in the data buffer, or the amount of space available in the data buffer will be sent in the acknowledgment after the printer receives a data string: • Filled - The total number of items in the buffer, including the last set of data that was received will be sent. • Empty - The total available buffer space, after the last set of data that was received will be sent. Default setting: Filled
Ack/Nack buffer format	Note This setting is only valid, if <i>Ack type</i> is set to <i>BufferDepth</i>. Select the format of the returning acknowledgment after each data string is received. • Text • 1 byte • 2 bytes (BE) • 2 bytes (LE) Default setting: Text
Ack schedule	Select when an acknowledgment is sent after a data string has been received: • Immediately - The acknowledgment will be sent immediately after the string has been received. • AfterParsing - The acknowledgment will be sent after the data has been checked and verified as ok. Default setting: AfterParsing
Persistence enabled	Tick (enable) the <i>Persistence enabled</i> tick box, to let the EDC elements in every label get the same persistent information from a single data string. This is used for "batch" printing. See, What is Batch Printing? on page 8. Un-tick (disable) the <i>Persistence enabled</i> tick box, to let the EDC elements in every label get unique information from different data strings in the EDC buffer. This is used for "Buffer" printing. See, What is Buffer Printing? on page 7. Note If the persistence enabled tick box is un-ticked (disabled), it is possible to run out of EDC data. The printer will use the queued data and will display a buffer empty alert when the data runs out. Default setting: Un-ticked (Disabled)

EDC SETUP

Setting Name	Description
History checked	Tick (enable) the <i>History checked</i> tick box, to let the printer check if the same EDC data has already been printed. To define the behavior of this setting, see How to Setup EDC History Queue on page 28. Default setting: Un-ticked (Disabled)
Maximum queue size	Define the maximum number of items that the data buffer queue can store. For example, if this value is set to 10, and 11 data strings are sent to the printer, the 11th data string will be rejected. Default setting: 32
Expiry entries enabled	Tick (enable) the <i>Expiry entries enabled</i> tick box, to delete items from the EDC buffer queue if they have not been printed after a specified time. The time allowed before deletion is set in the <i>Expiry entries lifetime (s)</i> setting. This setting is useful in the alcohol and tobacco industries. It can be used to ensure tax codes are removed from the printer's data queue if they have not been printed after a specified time. Default setting: Un-ticked (Disabled)
Expiry entries lifetime (s)	If <i>Expiry entries enabled</i> tick box is ticked (enabled), define the time in seconds that EDC data can stay in the data buffer queue before it is deleted. Default setting: 0
Store ID enabled	Tick (enable) the <i>Store ID enabled</i> tick box, to enable the return of data string IDs that are in the buffer when the list of expired entries is queried. The data string IDs are returned, instead of the data so that space and time are optimised. Default setting: Un-ticked (Disabled)
Exp. entries report format	If the <i>Store ID enabled</i> tick box is ticked (enabled), define the format of the expired entries report. Default setting: Empty
Expired ID length	If the <i>Store ID enabled</i> tick box is ticked (enabled), define the length of the data string identifier. The identifier will be part of the data string (the total length of the EDC string will be the ID length + Data length). Range: 0-5 Default setting: 0
Buffer low level	Set the buffer low level value. When the number of EDC strings in the buffer drops below this value an action defined by the <i>Buffer low level action</i> setting will occur. Default setting: 1

Setting Name	Description
Buffer low level action	Choose the action that will occur when the number of EDC strings in the buffer drops below the <i>Buffer low level</i> value. • None - No action will occur. • Send message - A message written by the user will be sent. • Pulse GPO - An output pin on the user port will be pulsed. To setup the output pin, see How to Setup Buffer High/Low Outputs on page 26. Default setting: None
Buffer low level message	If the <i>Buffer low level action</i> setting is set to <i>Send message</i>. Define the message that will be shown when buffer drops below the <i>Buffer low level</i> value. Default setting: Empty
Buffer high level	Set the buffer high level value. When the number of data strings in the buffer goes above this value an action defined by the <i>Buffer high level action</i> setting will occur. Default setting: 32
Buffer high level action	Choose the action when the number of data strings in the buffer goes above the <i>Buffer high level</i> value. • None - No action will occur. • Send message - A message written by the user will be sent. • Pulse GPO - An output pin on the user port will be pulsed. To setup the output pin, see How to Setup Buffer High/Low Outputs on page 26. Default setting: None
Buffer high level message	If the <i>Buffer high level action</i> setting is set to <i>Send message</i>. Define the message that will be shown when buffer goes above the <i>Buffer high level</i> value. Default setting: Empty

EDC SETUP

Setting Name	Description
Buffer empty behavior	Note If the buffer becomes empty, counter elements in the label and other statistics will continue to increment when print trigger signals are received, no matter which option below is selected. Choose the action when there is no data in the EDC buffer, but the label requires data. • Blank EDC element - The label will be printed, but the external data element will not be printed. • Blank label - The label will not be printed. • Default EDC element - The external data element's default value will be printed. Default setting: Blank EDC element
Buffer overwrite	Tick (enable) the <i>Buffer overwrite</i> to box to let data in the buffer be overwritten. This setting may be useful, if each product is printed with unique data but a product has been removed from the production line. This setting will let the data buffer be updated to make sure the correct data is printed on the correct product. To read a more detailed description of this feature, see Buffer Overwrite Notes on page 24. Default setting: Un-ticked (Disabled)

9. Tick (enable) the *Protocol enabled* tick box.

10. To save the new settings, select *Home > Setup > Production line setup > Power options > Restart* and wait for the printer to restart.

HOW TO SETUP AN EDC ETHERNET CONNECTION

Note If the settings described in this topic are changed when a label containing EDC elements is loaded for printing, re-load the label to print.

For the connection to be successful, it is essential that the settings described in this topic are set correctly and mirror the settings of the device that is used to transmit data to the printer.

To setup the printer's Ethernet settings:

1. Select *Home > Setup > Printer network > Ethernet > Edit*.
2. Two different actions are available for this step depending if the network uses static or dynamic IP (Internet Protocol) addresses:

Network Type	Required Action
Dynamic (IP addresses are assigned automatically by the network's DHCP (dynamic host connection protocol) server)	Make sure that the <i>Enabled DHCP</i> tick box is ticked (enabled).

Network Type	Required Action
Static (IP addresses are assigned manually)	Make sure that the <i>Enabled DHCP</i> tick box is un-ticked (disabled) and configure the settings described below: • IP address - Set the printer's IP address. Get an IP address from the network administrator and verify that the IP address is not used by another other device on the network. Note The first 3 sets of digits in the IP address must match other devices on the network that the printer is connected to. The fourth set of digits must be unique. For example, if the device that will transmit EDC data to the printer is connected to the same network and has the IP address 192.168.1.5, set the printer's IP address to 192.168.1.55 (if that address is not already in use). • Subnet mask - Set the subnet mask. The subnet mask must match other devices on the same network. • Default gateway - If the printer needs to communicate with devices on a different network (or the internet), enter the IP address of the network's default gateway (router).

3. If any settings were changed, select *Home > Setup > Production line setup > Power options > Restart* and wait for the printer to restart.

To setup and enable the EDC connection via Ethernet:

1. Select *Home > Setup > Printer network > Protocol settings*.
2. Select the *Protocol Setting* drop down setting and select *EDC TCP*.
3. Un-tick (disable) the *Protocol enabled* tick box.

Note The settings cannot be changed, when the protocol is enabled.

4. EDC settings described in the table below can now be defined. After these settings have been defined, tick (enable) the *Protocol Enabled* tick box.

Setting Name	Description
TCP Port	The port number for the printer's Ethernet Port: Default setting: 16000
Encoding	Set the data string's encoding method: • ASCII • UTF-8 • Unicode(BE) • Unicode(LE) Default setting: ASCII

Setting Name	Description
Start marker	Define the start marker that is used to mark the start of printable items in the data string. Default setting: <STX>
End marker	Define the end marker that is used to mark the end of printable items in the data string. Default setting: <ETX>
Packet length	If <i>Data packet type</i> is set to <i>Use specific length</i> this setting defines the length of printable items in the data string that the printer will look for. The length is defined in bytes. Default setting: 1
Data packet type	Set if the start and end of each data string will be marked with start and end markers. Or, set if the printer will look for items of a specific length. Default setting: Use start end marker
Ack type	Setup the acknowledgment that the printer will send after the printer receives a data string: • None - No acknowledgement will be sent. • Default - The letters ACK (41 43 4B) will be sent. • RegExpression - A user defined acknowledgment will be sent (This feature will be enabled in a later software update). • DefaultSpecialChars - ACK (06) will be sent. • BufferDepth - The number of items in the data buffer, or the amount of space in the data buffer will be sent. This setting only works for a buffer size up to 255, otherwise the responses will "loop". The behavior is defined by the <i>Ack/Nack buffer depth</i> setting. Default setting: DefaultSpecialChars
Ack/Nack buffer depth	Note This setting is only valid, if <i>Ack type</i> is set to <i>BufferDepth</i> . Select if the number of items in the data buffer, or the amount of space available in the data buffer will be sent in the acknowledgment after the printer receives a data string: • Filled - The total number of items in the buffer, including the last set of data that was received will be sent. • Empty - The total available buffer space, after the last set of data that was received will be sent. Default setting: Filled

Setting Name	Description
Ack/Nack buffer format	Note This setting is only valid, if <i>Ack type</i> is set to <i>BufferDepth</i>. Select the format of the returning acknowledgment after each data string is received. • Text • 1 byte • 2 bytes (BE) • 2 bytes (LE) Default setting: Text
Ack schedule	Select when an acknowledgment is sent after a data string has been received: • Immediately - The acknowledgment will be sent immediately after the string has been received. • AfterParsing - The acknowledgment will be sent after the data has been checked and verified as ok. Default setting: AfterParsing
Persistence enabled	Tick (enable) the <i>Persistence enabled</i> tick box, to let the EDC elements in every label get the same persistent information from a single data string. This is used for "batch" printing. See, What is Batch Printing? on page 8. Un-tick (disable) the <i>Persistence enabled</i> tick box, to let the EDC elements in every label get unique information from different data strings in the EDC buffer. This is used for "Buffer" printing. See, What is Buffer Printing? on page 7. Note If the persistence enabled tick box is un-ticked (disabled), it is possible to run out of EDC data. The printer will use the queued data and will display a buffer empty alert when the data runs out. Default setting: Un-ticked (Disabled)
History checked	Tick (enable) the <i>History checked</i> tick box, to let the printer check if the same EDC data has already been printed. To define the behavior of this setting, see How to Setup EDC History Queue on page 28. Default setting: Un-ticked (Disabled)
Maximum queue size	Define the maximum number of items that the data buffer queue can store. For example, if this value is set to 10, and 11 data strings are sent to the printer, the 11th data string will be rejected. Default setting: 32

EDC SETUP

Setting Name	Description
Expiry entries enabled	Tick (enable) the <i>Expiry entries enabled</i> tick box, to delete items from the EDC buffer queue if they have not been printed after a specified time. The time allowed before deletion is set in the <i>Expiry entries lifetime (s)</i> setting. This setting is useful in the alcohol and tobacco industries. It can be used to ensure tax codes are removed from the printer's data queue if they have not been printed after a specified time. Default setting: Un-ticked (Disabled)
Expiry entries lifetime (s)	If <i>Expiry entries enabled</i> tick box is ticked (enabled), define the time in seconds that EDC data can stay in the data buffer queue before it is deleted. Default setting: 0
Store ID enabled	Tick (enable) the <i>Store ID enabled</i> tick box, to enable the return of data string IDs that are in the buffer when the list of expired entries is queried. The data string IDs are returned, instead of the data so that space and time are optimised. Default setting: Un-ticked (Disabled)
Exp. entries report format	If the <i>Store ID enabled</i> tick box is ticked (enabled), define the format of the expired entries report. Default setting: Empty
Expired ID length	If the <i>Store ID enabled</i> tick box is ticked (enabled), define the length of the data string identifier. The identifier will be part of the data string (the total length of the EDC string will be the ID length + Data length). Range: 0-5 Default setting: 0
Buffer low level	Set the buffer low level value. When the number of EDC strings in the buffer drops below this value an action defined by the <i>Buffer low level action</i> setting will occur. Default setting: 1
Buffer low level action	Choose the action that will occur when the number of EDC strings in the buffer drops below the <i>Buffer low level</i> value. • None - No action will occur. • Send message - A message written by the user will be sent. • Pulse GPO - An output pin on the user port will be pulsed. To setup the output pin, see How to Setup Buffer High/Low Outputs on page 26. Default setting: None
Buffer low level message	If the <i>Buffer low level action</i> setting is set to <i>Send message</i>. Define the message that will be shown when buffer drops below the <i>Buffer low level</i> value. Default setting: Empty

Setting Name	Description
Buffer high level	Set the buffer high level value. When the number of data strings in the buffer goes above this value an action defined by the <i>Buffer high level action</i> setting will occur. Default setting: 32
Buffer high level action	Choose the action when the number of data strings in the buffer goes above the <i>Buffer high level</i> value. • None - No action will occur. • Send message - A message written by the user will be sent. • Pulse GPO - An output pin on the user port will be pulsed. To setup the output pin, see How to Setup Buffer High/Low Outputs on page 26. Default setting: None
Buffer high level message	If the <i>Buffer high level action</i> setting is set to <i>Send message</i> . Define the message that will be shown when buffer goes above the <i>Buffer high level</i> value. Default setting: Blank
Buffer empty behavior	Note If the buffer becomes empty, counter elements in the label and other statistics will continue to increment when print trigger signals are received, no matter which option below is selected. Choose the action when there is no data in the EDC buffer, but the label requires data. • Blank EDC element - The label will be printed, but the external data element will not be printed. • Blank label - The label will not be printed. • Default EDC element - The external data element's default value will be printed. Default setting: Blank EDC element
Buffer overwrite	Tick (enable) the <i>Buffer overwrite</i> to box to let data in the buffer be overwritten. This setting may be useful, if each product is printed with unique data but a product has been removed from the production line. This setting will let the data buffer be updated to make sure the correct data is printed on the correct product. To read a more detailed description of this feature, see Buffer Overwrite Notes on page 24. Default setting: Un-ticked (Disabled)

5. Tick (enable) the *Protocol enabled* tick box.
6. To save the new settings, select *Home > Setup > Production line setup > Power options > Restart* and wait for the printer to restart.

BUFFER OVERWRITE NOTES

Buffer overwrite is useful in the example described below.

Parts coming down a production line may look like this:

Products:	O	O	O	O	O	O	O	O	O	O	O	O	O	O
Conveyor:	-----													
Data Sent:	A	B	C	D	E	F	G	H	I	J	K	L	M	N
Data Printed:	A	B	C	D	E	F	G		H	J	K	L	M	N

Part H is missing because it has been removed from the production line. If the printer is sent data for the missing part, it's not possible to send the correct unique data for the next part because the EDC queue will be full. Therefore, the product after the gap will have the wrong data printed on it.

However, if Buffer Overwrite is enabled, the customer can overwrite the buffer.

Now the same production run can be corrected:

Products:	O	O	O	O	O	O	O	O	O	O	O	O	O	O
Conveyor:	-----													
Data Sent:	A	B	C	D	E	F	G	H	I	J	K	L	M	N
Data Printed:	A	B	C	D	E	F	G		I	J	K	L	M	N

Buffer overwrite may not be applicable to most applications as it limits the queue depth to 1 (so the customer doesn't need to think about which item to overwrite). However, for other applications such as electronics parts marking, this feature is useful.

HOW TO SETUP THE BUFFER EMPTY ALERT

Note These settings were added in software version 01.40.0915.

The printer will display an alert, when the EDC data buffer becomes empty. The behavior of this alert is configurable.

The severity of the EDC buffer empty alert can be set to:

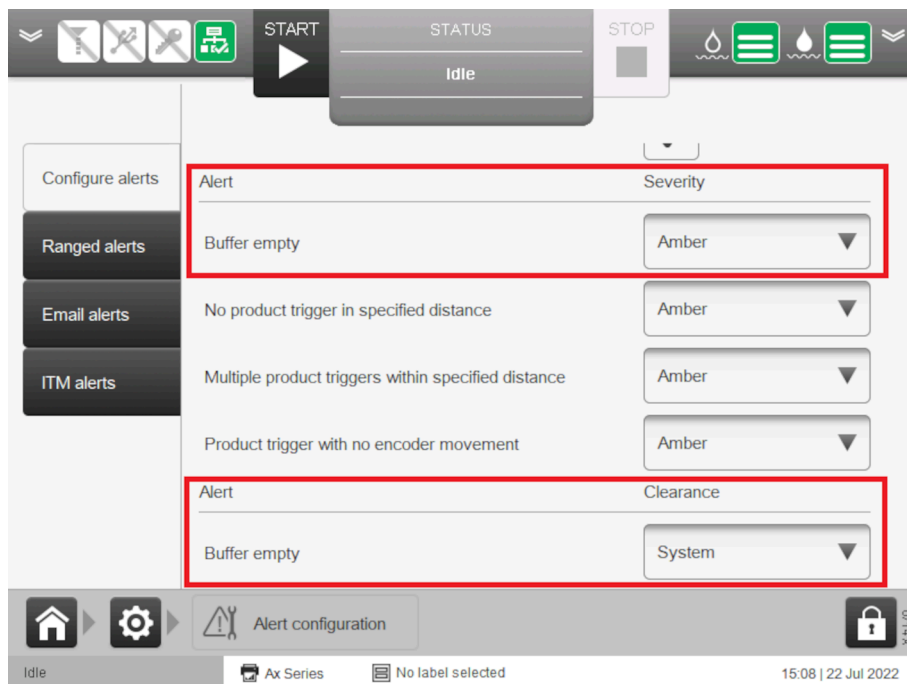
- **Amber** - Printing will continue. An amber colored alert will be displayed on the printer's user interface. If a beacon is connected to the printer, the amber light will illuminate.
- **Red** - The ink jet will continue running, although the printer will not make another print until the alert is cleared. A red colored alert will be displayed on the printer's user interface. If a beacon is connected to the printer, the red light will illuminate.

The clearance level for the EDC buffer empty alert can be set to:

- **System** - The alert can only be cleared by the system i.e. when the EDC buffer is refilled.
- **User** - The alert can be cleared by a user/operator in the printer's user interface.
- **User Acknowledge** - The alert can only be acknowledged, but not cleared by a user. The alert can only be cleared by the system.

To change the settings for the EDC buffer empty alert:

1. On the TouchScreen, select *Home > Setup > Alert configuration > Configure alerts*.
2. Change the *Severity* and *Clearance* drop down boxes for *Buffer empty*.



HOW TO SETUP BUFFER HIGH/LOW OUTPUTS

Note To use this feature, the GPIO or Extended IO Pack must be installed on the printer.

When the number of items in the data buffer meets a user defined high or low value, an output from a pin on one of the printer's IO sockets can be triggered. For example, the outputs could be setup to ask for more data, or request that no more data is sent.

To enable this feature:

1. Select *Home > Setup > Printer network > Protocol settings*.
2. Select the *Protocol Setting* drop down and select *EDC Serial* or *EDC TCP*.
3. Un-tick (disable) the *Protocol enabled* tick box.

Note The settings cannot be changed, when the protocol is enabled.

4. Adjust the settings listed below:

Setting Name	Description
Buffer low level	Set the buffer low level value. When the number of EDC strings in the buffer drops below this value, the output will be triggered.
Buffer low level action	Select <i>Pulse GPO</i> .
Buffer high level	Set the buffer high level value. When the number of data packets in the buffer goes above this value the output will be triggered.
Buffer high level action	Select <i>Pulse GPO</i> .

5. Tick (enable) the *Protocol enabled* tick box.
6. To save the new settings, select *Home > Setup > Production line setup > Power options > Restart* and wait for the printer to restart.

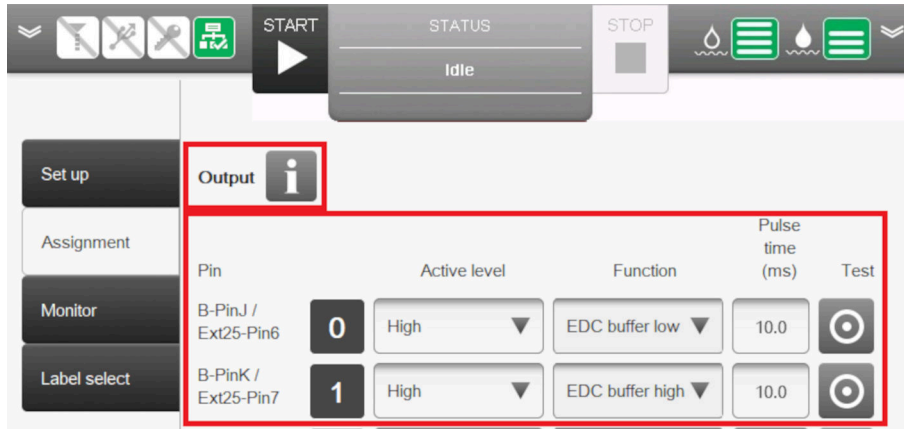
To select the I/O socket and pin that will be triggered:

1. Select *Home > Setup > I/O port > Assignment*.
2. Scroll down to *Output*, set the function of 1 output pin to *EDC buffer low* and another output pin to *EDC buffer high*.

EDC SETUP

- Set the *Active level*, and *Pulse time (ms)* values for the 2 outputs.

Note The first 4 outputs shown on this screen are for the printer's GPIO or Extended IO 25-way socket. The remaining outputs are used for the Extended IO 37-way socket.



HOW TO SETUP EDC HISTORY QUEUE

External data strings sent to the printer can be saved in a history queue. The history queue lets the printer check if duplicate data is received and to raise an alert if required.

To setup the EDC history queue settings:

1. Select *Home > Setup > I/O Port > Setup*.
2. Scroll down to the *Other* heading.
3. Configure the settings described below:

Setting Name	Description
EDC history queue depth	This setting defines the maximum number of data strings that can be saved in the history queue. Range: 0 - 2048 Default setting: 1024
EDC history queue maximum duplicates	Define the number of duplicate data strings allowed before a printer alert is triggered. For example, if this setting is set to 10, then 10 duplicate data strings will be allowed before the alert is triggered. Range: 0 - 100 Default setting: 2
EDC history queue full action	Define the printer's behavior when the EDC history queue becomes full: • Reset - Clear the history queue and start again. • Overwrite - Overwrite the history queue. Default setting: Reset
EDC history queue preserve over power off	Tick (enable) the <i>EDC history queue preserve over power off</i> tick box to preserve the EDC history queue when the printer is powered off. If the tick box is un-ticked (disabled), the EDC history queue will be deleted when the printer is powered off. Default setting: Un-ticked (disabled)

HOW TO VIEW AND RESET THE DATA QUEUE & STATISTICS

The number of strings in the data buffer queue and statistics about the data buffer queue can be viewed and reset on the printer.

To view and reset the data buffer queue and data buffer queue statistics:

1. Select *Home > Setup > I/O Port > Monitor*.
2. The settings described below can now be changed:

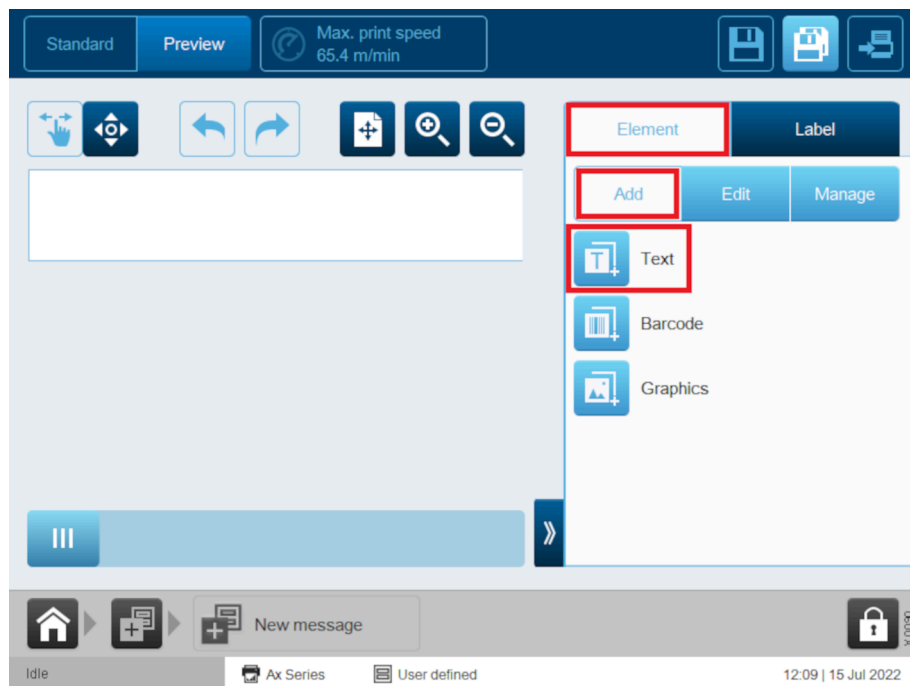
Setting Name	Description
'EdcSerial' queue state	Displays the number of data strings transmitted over serial connection that are stored in the EDC buffer queue. Select <i>Clear</i> to clear the EDC buffer queue.
'EdcTcpSing' queue state	Displays the number of data strings transmitted over Ethernet TCP/IP connection that are stored in the EDC buffer queue. Select <i>Clear</i> to clear the EDC buffer queue.
EDC history reset	Data strings received by the printer can be stored in a history log. This allows new data to be checked against historic data to avoid printing duplicate information. This setting displays the number of EDC history logs stored in the printer. Select <i>Clear</i> to clear the EDC history log.

HOW TO CREATE AND ADD AN EDC TEXT ELEMENT TO A LABEL

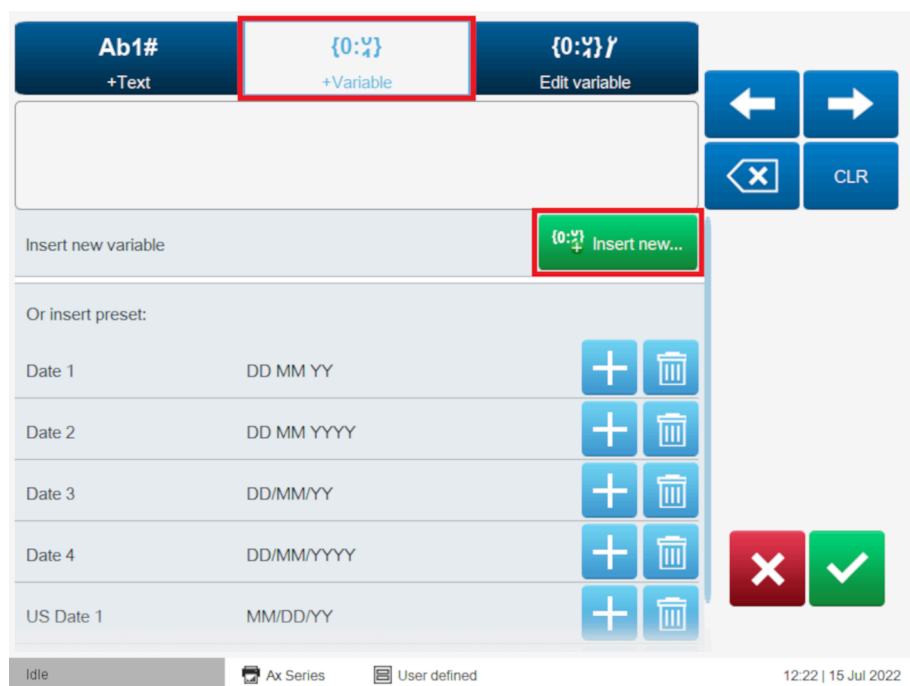
To print external data, an external data element needs to be added to the label.

To add an external data element into the label design:

1. In the label creator, select *Element > Add > Text*.



2. Select *+Variable > Insert new...*



3. Select *External data*.

4. The external data settings can now be defined:

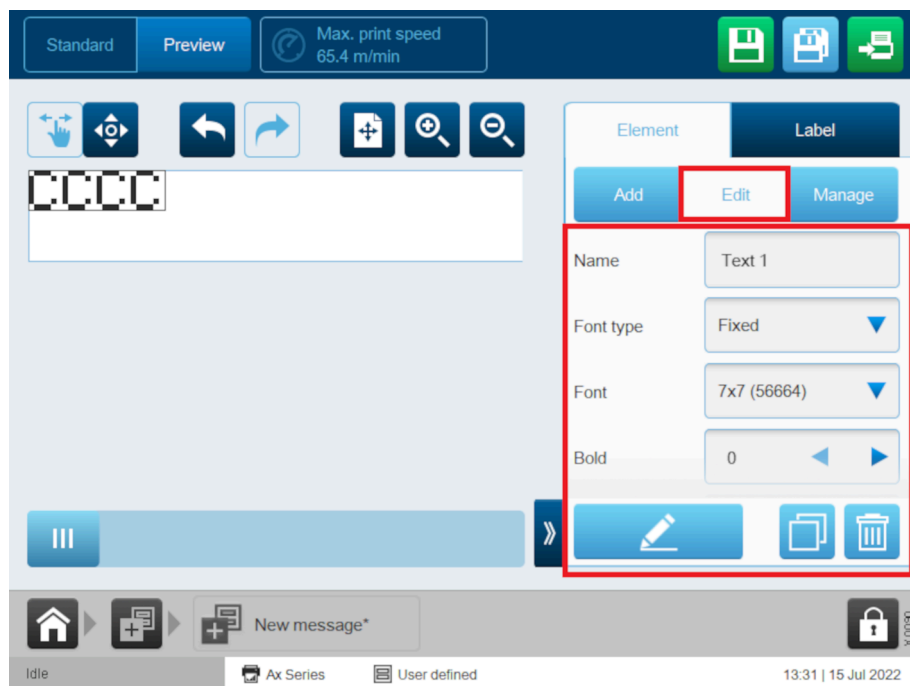
Setting Name	Description
Source	Select the external data communication source: • EDC TCP - EDC data is transmitted to the printer via Ethernet. • EDC Serial - EDC data is transmitted to the printer via RS232.
Length	Set the maximum length of the printed data. For example, if the data received by the printer is 01234 and <i>Length</i> is set to 3, the printed data will be 012.
Offset	The <i>Offset</i> value specifies which part of the data will be printed. For example, if the data received by the printer is 01234, <i>Length</i> is set to 3 and <i>Offset</i> is set to 2, the printed data will be 234.
Delimited	Tick (enable) the <i>Delimited tick</i> box to display the <i>Delimiter</i> and <i>Index</i> settings.
Delimiter	Note This setting is only visible, if the <i>Delimited tick</i> box is ticked (enabled). If data sent to the printer is made up of multiple data blocks, the delimiter is the character that separates each data block. The <i>Delimiter</i> setting specifies the delimiter character used in the data stream that is sent to the printer. The delimiter can be any character.
Index	Note This setting is only visible, if the <i>Delimited tick</i> box is ticked (enabled). The <i>Index</i> value selects a specific block in the data stream to print. The first block in the data stream is index 0. For example, if the data stream is 123,456,789,132,425 and <i>Index</i> is set to 0, the 1st data block will be printed which is 123. Whereas, if <i>Index</i> is set to 3, the 4th data block will be printed which is 132.

Setting Name	Description
Example data	Note This setting does not apply to the data that will be printed. This setting applies to the label preview on the user interface only. Choose how data in the label preview will be shown: • 0 - Only show numeric character types in the label preview. • L - Only show alpha character types (upper and lower case) in the label preview. • A - Only show alpha numeric character types (upper and lower case) in the label preview. • C - Show all character types in the label preview. • M - Only show upper case alpha character types in the label preview. • m - Only show lower case alpha character types in the label preview. • Z - Only show upper case alpha numeric character types in the label preview. • z - Only show lower case alpha numeric character types in the label preview.
Default value	Set a value that can be printed in place of missing EDC data. To use this feature, the <i>Buffer empty behavior</i> setting (Found in: <i>Home > Setup > Printer network > Protocol settings</i>) must be set to <i>Default EDC element</i>.

5. Select the green *Tick* icon to confirm the EDC element settings.
6. Select the green *Tick* icon to add the EDC element to the label design.

EDC SETUP

7. Select *Edit* and use the settings on the right side of the screen to change the EDC element's appearance.



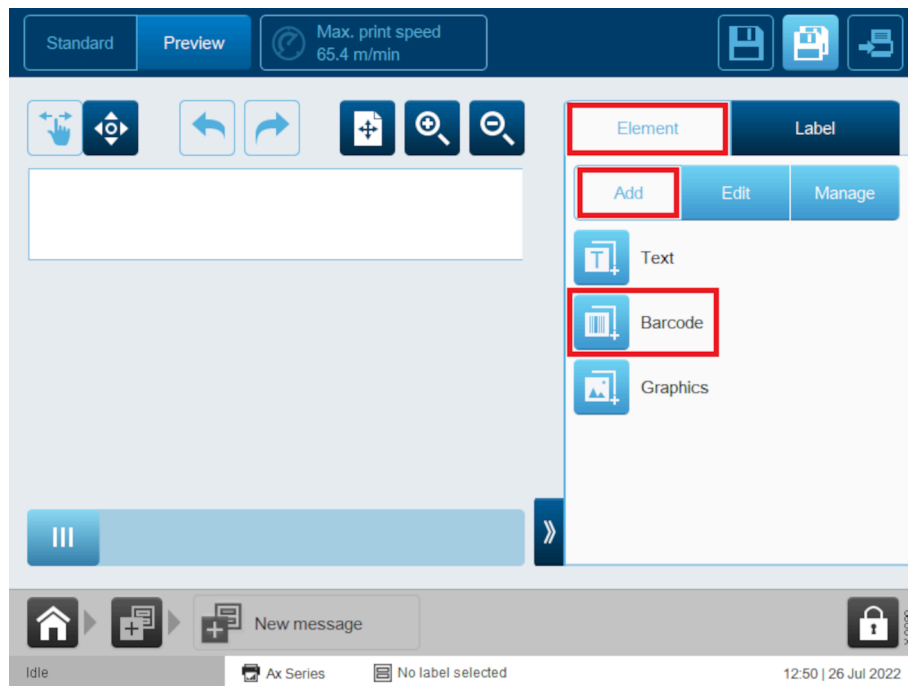
8. Select the Save icon to save the change.

HOW TO CREATE AND ADD AN EDC BARCODE ELEMENT TO A LABEL

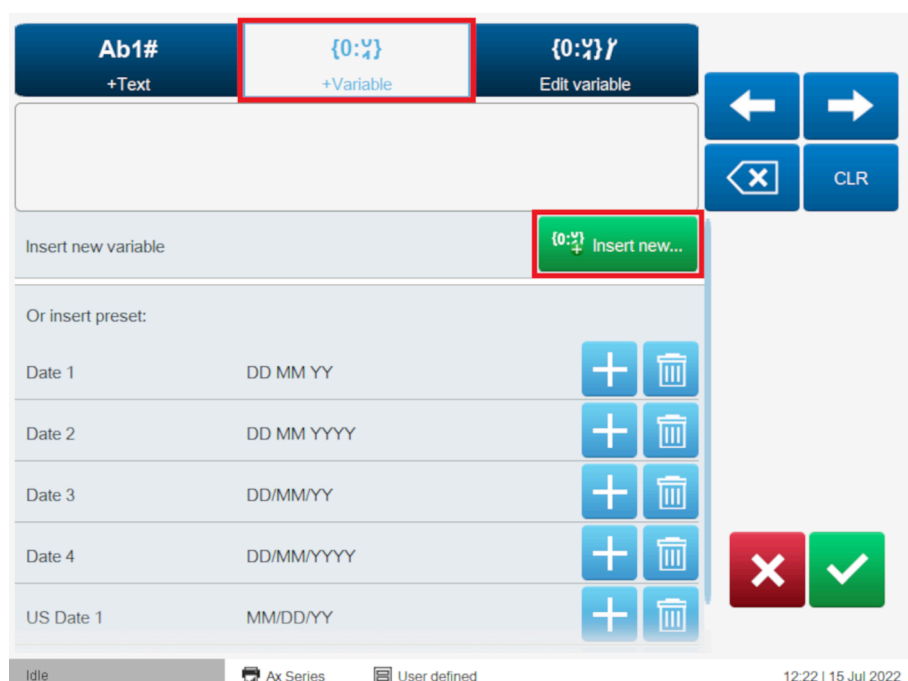
To print external data, an external data element needs to be added to the label. External data elements can be placed inside Barcode elements.

To add an external data element within a barcode into the label design:

1. In the label creator, select *Element > Add > Barcode*.



2. Select the *Type* drop down setting, select the required barcode type and select *Add*.
3. Select *+Variable > Insert new...*



4. Select *External data*.

5. The external data settings can now be defined:

Setting Name	Description
Source	Select the external data communication source: • EDC TCP - EDC data is transmitted to the printer via Ethernet. • EDC Serial - EDC data is transmitted to the printer via RS232.
Length	Set the maximum length of the printed data. For example, if the data received by the printer is 01234 and <i>Length</i> is set to 3, the printed data will be 012.
Offset	The <i>Offset</i> value specifies which part of the data will be printed. For example, if the data received by the printer is 01234, <i>Length</i> is set to 3 and <i>Offset</i> is set to 2, the printed data will be 234.
Delimited	Tick (enable) the <i>Delimited tick</i> box to display the <i>Delimiter</i> and <i>Index</i> settings.
Delimiter	Note This setting is only visible, if the <i>Delimited tick</i> box is ticked (enabled). If data sent to the printer is made up of multiple data blocks, the delimiter is the character that separates each data block. The <i>Delimiter</i> setting specifies the delimiter character used in the data stream that is sent to the printer. The delimiter can be any character.
Index	Note This setting is only visible, if the <i>Delimited tick</i> box is ticked (enabled). The <i>Index</i> value selects a specific block in the data stream to print. The first block in the data stream is index 0. For example, if the data stream is 123,456,789,132,425 and <i>Index</i> is set to 0, the 1st data block will be printed which is 123. Whereas, if <i>Index</i> is set to 3, the 4th data block will be printed which is 132.

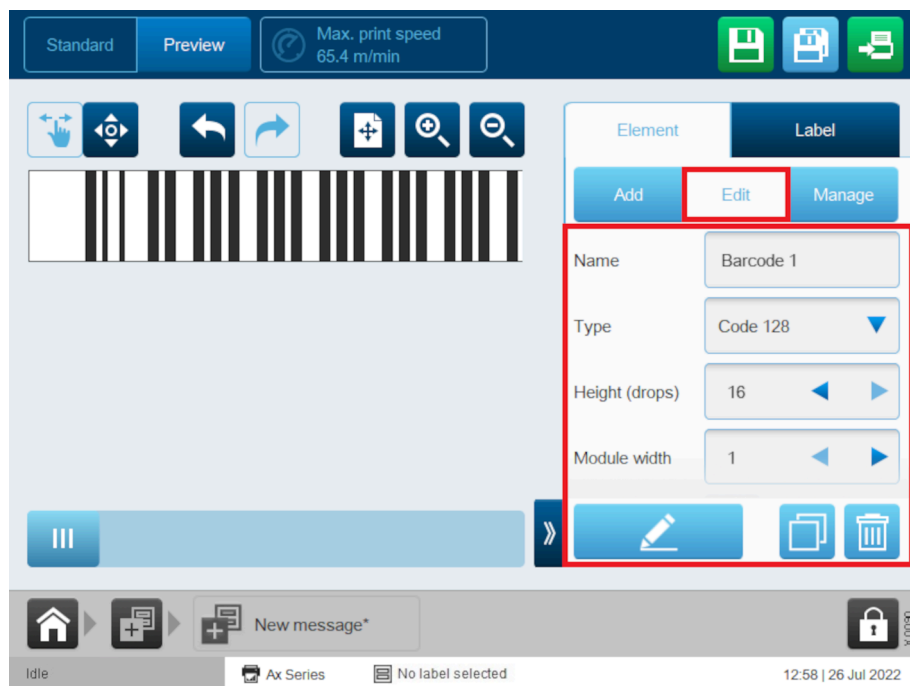
Setting Name	Description
Example data	Notes 1. This setting does not apply to the data that will be printed. This setting applies to the label preview on the user interface only. 2. When an EDC element is added to a barcode, the worst case scenario is assumed for barcode sizing. For example: • If the external data variable has a length of 10 in a Datamatrix code with example data of CCCCCCCCCC, it will allow a minimum size code of 16x16. • If the external data variable has a length of 10 in a Datamatrix code with example data of 0000000000, it will allow a minimum size code of 12x12. Choose how data in the label preview will be shown: • 0 - Only show numeric character types in the label preview. • L - Only show alpha character types (upper and lower case) in the label preview. • A - Only show alpha numeric character types (upper and lower case) in the label preview. • C - Show all character types in the label preview. • M - Only show upper case alpha character types in the label preview. • m - Only show lower case alpha character types in the label preview. • Z - Only show upper case alpha numeric character types in the label preview. • z - Only show lower case alpha numeric character types in the label preview.
Default value	Set a value that can be printed in place of missing EDC data. To use this feature, the <i>Buffer empty behavior</i> setting (Found in: <i>Home > Setup > Printer network > Protocol settings</i>) must be set to <i>Default EDC element</i>.

6. Select the green *Tick* icon to confirm the element settings.

7. Select the green *Tick* icon to add the element to the label design.

EDC SETUP

8. Select *Edit* and use the settings on the right side of the screen to change the element's appearance.



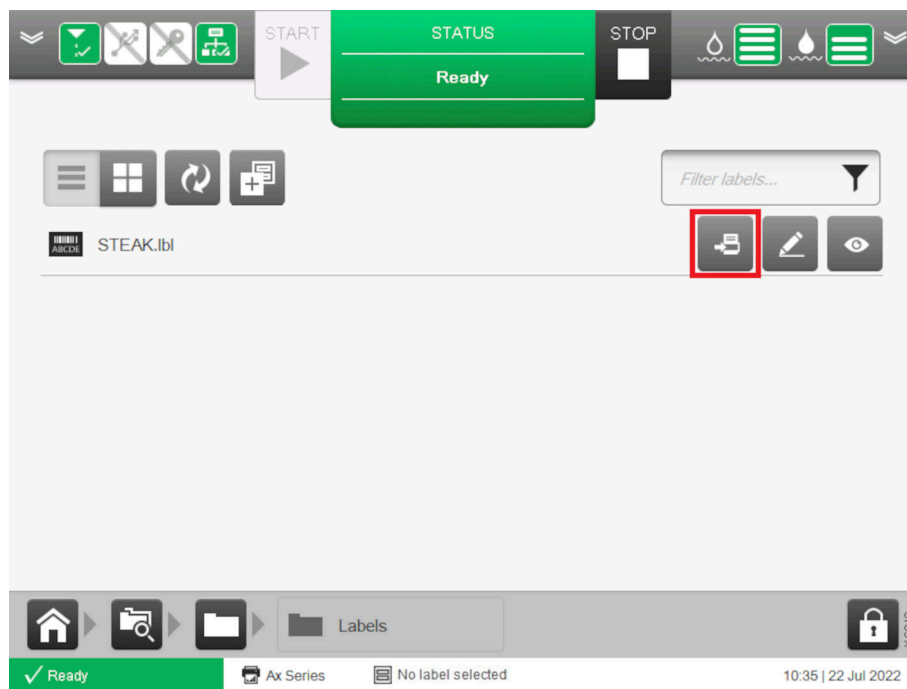
9. Select the Save icon to save the change.

HOW TO PRINT A LABEL WITH EDC ELEMENTS

To print a label with EDC elements:

1. Make sure the printer is in the *Ready* state. If the printer is not in the *Ready* state, select *START* at the top of the screen.
2. From the *Home* screen, select *Label finder*.
3. Open the *Labels* folder, find the label and select the *Send to print* icon.

Note If the protocol settings are ever changed, the label will need to be sent to print again.

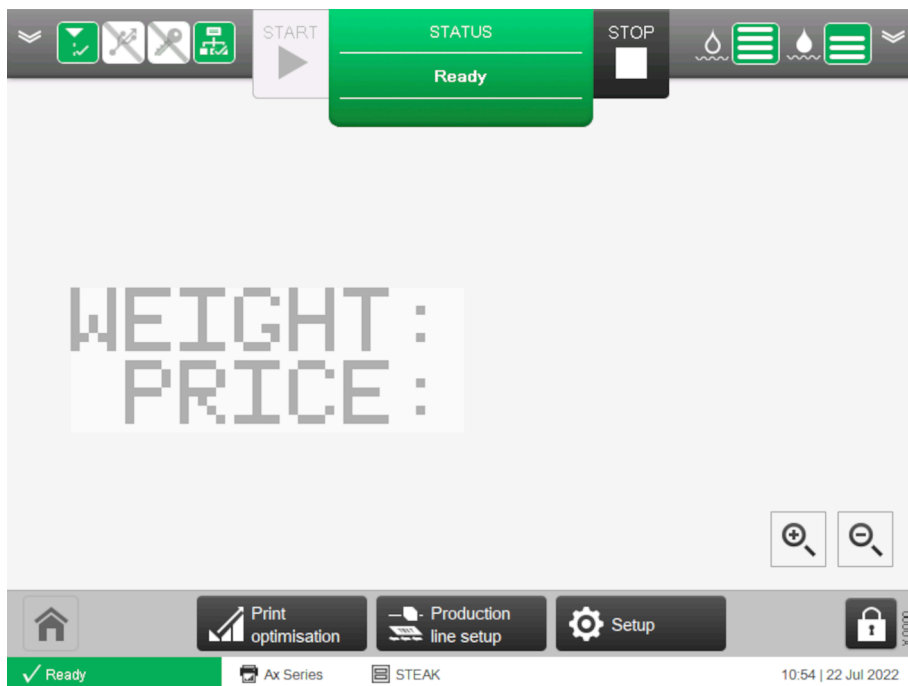


- Select the background of the user interface to preview the label.

Note The name of the loaded label is displayed at the bottom of the screen.

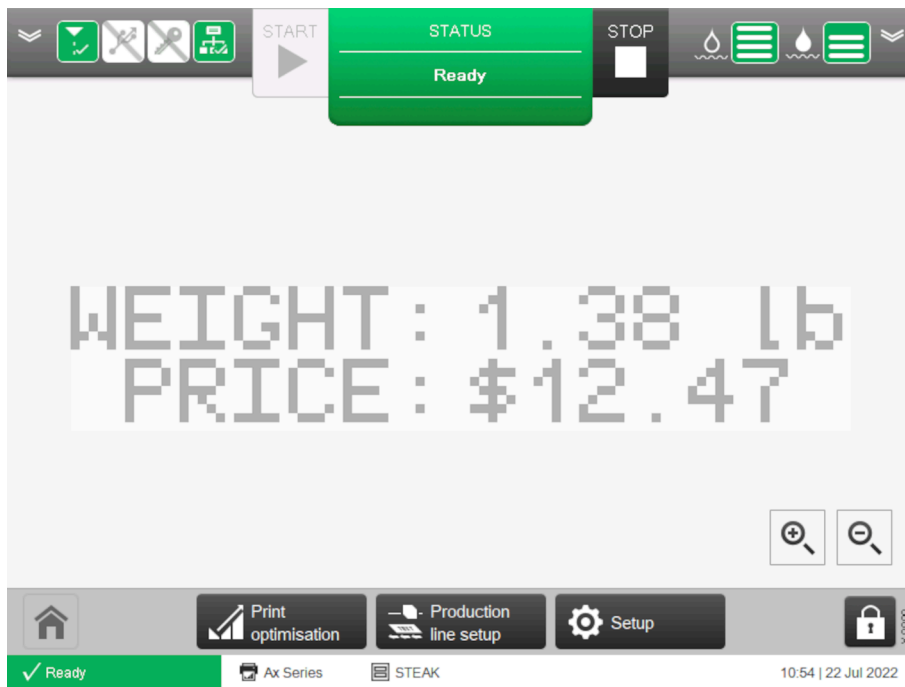


- EDC elements in the label are blank, until both the EDC data and a print trigger signal have been received by the printer.



EDC SETUP

6. Send the EDC data to the printer and trigger the product detect sensor to begin printing to see the label preview update.



PART 3 USING EDC WITH CODENET

EDC WITH CODENET INTRODUCTION

Codenet protocol can also be used to select and load a label to print, to populate the EDC buffer, and to clear the EDC buffer/history queue.

If RS232 (Serial) is used by 1 communication protocol, the socket cannot be used simultaneously for any other communication protocol. It will not be possible to use Codenet and EDC via the same serial communication socket. One protocol will need to use RS232 (Serial) and the other will need to use Ethernet (TCP). Or, both communication protocols can use Ethernet (TCP) simultaneously with no problem.

HOW TO SETUP A CODENET RS232 CONNECTION

Note If the settings described in this topic are changed when a label containing EDC elements is loaded for printing, re-load the label to print.

For the connection to be successful, it is essential that the settings described in this topic are set correctly and mirror the settings of the device that is used to transmit data to the printer.

To setup and enable the EDC connection via RS232:

1. Select *Home > Setup > Printer network > Protocol settings*.
2. Select the *Protocol Setting* drop down setting and select *Codenet*.
3. Un-tick (disable) the *Protocol enabled* tick box.
4. Select the *Protocol mode* drop down setting and select *Serial*.

Note The settings cannot be changed, when the protocol is enabled.

5. Select the *Serial Com* drop down setting and select either:

Serial Com Setting	Description
COM1	For communication via the RS232 Pack's RS232 Socket (if fitted).
COM2	For communication via the Extended IO Pack's 37-Way Socket (if fitted).

6. Select *Advanced*.
7. The settings described in the table below can be set using the drop down boxes:

Setting Name	Description
Baud rate	Select the baud rate between the printer and the device that the printer is connected to. Range: 1200-115200 Default setting: 9600

Setting Name	Description
Data bits	Select the number of data bits in each character. Range: 7-8 Default setting: 8
Parity	Select the type of parity bit that will be expected. A parity bit is an extra data bit that will be sent with each data character to detect errors in the RS232 transmission. No parity - No parity bit will be expected. Odd - An odd parity bit will be expected. Even - An even parity bit will be expected. Mark - The parity bit will always be 1 but will not be used. Space - The parity bit will always be 0 but will not be used. Default setting: No parity
Stop bits	Select the number of stop bits that will be sent at the end of each character to enable the detection of the end of the character. Valid settings: 1, 1.5, or 2 Default setting: 1

Setting Name	Description
Flow control	Select the type of handshake signal that will be used to pause and resume the transmission of data. None - Flow control will not be enabled. None, DTR enabled - - The DTR output line will be asserted (+12V) whenever the printer is ready to receive data. - The DTR output line will be de-asserted (-12V) whenever the printer is not ready for data entry. DTR/DSR - - The DTR output line will be asserted (+12V) whenever the printer is ready to receive data. - The DTR output line will be de-asserted (-12V) whenever the printer is not ready for data entry. - Whenever the DSR input line is asserted (+12V), the printer will be enabled for the sending of data. - Whenever the DSR input line is de-asserted (-12V), the printer will suspend data output. RTS/CTS - - The RTS output line will be asserted (+12V) whenever the printer is ready to receive data. - The RTS output line will be de-asserted (-12V) whenever the printer is not ready for data entry. - Whenever the CTS input line is asserted (+12V), the printer will be enabled for the sending of data. - Whenever the CTS input line is de-asserted (-12V), the printer will suspend data output. XOn/XOff - Note When XOn/Xoff is selected, CTS and RTS signals must not be connected. - The printer will send a single ASCII <XOFF> (013) character whenever it requires data entry to be suspended. - The printer will send a single ASCII (011) character whenever it becomes ready to receive data. - Whenever the printer receives an <XOFF> character it will suspend its data output. Transmission from the printer will be enabled whenever it receives an character. Default setting: None

8. Select *Save*.

9. Settings described in the table below can now be defined. After these settings have been defined, tick (enable) the *Protocol Enabled* tick box.

Setting Name	Description
TCP Port	Define the port that will be used for codenet communication. Default setting: 7000
Response length	Define the length of the response data package: • Variable - ACK is one byte (0616), and NAK is four bytes (15, A, B, C). Where A, B and C represent a three character Hex error code. • Fixed - Both ACK and NAK are four bytes (06303030 and 15 A B C). Where A, B and C represent a three character Hex error code. Default setting: Variable
Send response	Define when a response to a data packet will be sent from the printer. • On Processed - The ACK or NAK will be sent when the printer has processed a data packet. • On Received - An ACK will be sent as soon as the printer receives a data packet. Note, the data has not yet been validated as correct. Default setting: On processed
LED state	This setting lets the LEDs on the printer cabinet to mimic the behaviour of different printer types. This is useful if a different printer type is replaced with an Ax-Series printer. • Standard - The LEDs on the printer cabinet will mimic the behavior of A-Series i-Tech and A-Series plus printers. • Green + Flash - The LEDs on the printer cabinet will mimic the behavior of an A-Series classic printer. Default setting: Standard

Setting Name	Description
Printer ID	This setting allows the printer to respond using the ID of older printer types. This is useful if an older printer type is replaced with an Ax-Series printer, as the controller may be expecting a different response value. • 00 = Codebox • 01 = Solo • 02 = Solo Twin Head • 03 = A300 • 04-19 = Not Allocated • 20 = Macrojet • 21 = Casecoder • 22 = A-Series plus A100 • 23 = A-Series plus A300 • 24 = A-Series plus Duo • 25-29 = Not Allocated • 30 = Ax-Series • 31 - 99 = Not Allocated
Label format	Select which label format/raster will be used when a data packet containing a label design is sent to the printer.

Setting Name	Description
Backward compatible	This tick box changes the printer behaviour if the 0x52 ('R' - Clear all labels) command is used in between changing labels during continuous printing. Important: If the gap between prints is important, it is recommended to not use the 'R' (Clear all labels) command in between changing labels during continuous printing. Using this command introduces the risk that the gap between the last print of the old label and the first print of the new label will change. Backward Compatible Un-ticked (Disabled) - A print trigger is needed to re-start printing after the new continuous print label is put online: (1) Printer is in the "Ready" state. (2) Select and put online a label that has the "Repeat mode" set to "Continuous", send the command: 'P' (Put label with 3 digit name from store online) or 'ON' (Put label with variable length name from store online) (3) Send command: 'N' (Software Print Go) Note Printing starts. (4) Send command: 'R' (Clear all labels) Note A blank "NONE Continuous Printing Label" is put online which causes printing to stop. (5) Select and put online a different label that has the "Repeat mode" set to "Continuous", send the command: 'P' (Put label with 3 digit name from store online) or 'ON' (Put label with variable length name from store online) (6) The new label is not printed until the 'N' (Software Print Go) command is sent again. Backward Compatible Ticked (Enabled) - Printing will start as soon as the new continuous print label is put online: (1) Printer is in the "Ready" state. (2) Select and put online a label that has the "Repeat mode" set to "Continuous", send the command: 'P' (Put label with 3 digit name from store online) or 'ON' (Put label with variable length name from store online) (3) To start printing, send the command: 'N' (Software Print Go) (4) Send command: 'R' (Clear all labels) Note A blank "NONE Continuous Printing Label" is put online which causes printing to stop. (5) Select and put online a different label that has the "Repeat mode" set to "Continuous", send the command: 'P' (Put label with 3 digit name from store online) or 'ON' (Put label with variable length name from store online)

Setting Name	Description
Backward compatible (Continued)	Printing starts as soon as the new label is put online. The command 'N' (Software Print Go) is not needed to start printing. Default setting: Variable
Counter persistence on	Note This setting was added in software version 01.40.0629 When this tick box is selected, the values of counters in any labels newly created through Codenet will not be reset to the start value each time the label is sent to print. For example: (1) The counter value in the label reaches 30 and the label is taken offline. (2) The label is sent to print again, the counter value will advance to 31 Default setting: Ticked (Enabled)

10. Tick (enable) the *Protocol enabled* tick box.

11. To save the new settings, select *Home > Setup > Production line setup > Power options > Restart* and wait for the printer to restart.

HOW TO SETUP A CODENET ETHERNET CONNECTION

Note If the settings described in this topic are changed when a label containing EDC elements is loaded for printing, re-load the label to print.

For the connection to be successful, it is essential that the settings described in this topic are set correctly and mirror the settings of the device that is used to transmit data to the printer.

To setup the printer's Ethernet settings:

1. Select *Home > Setup > Printer network > Ethernet > Edit*.
2. Two different actions are available for this step depending if the network uses static or dynamic IP (Internet Protocol) addresses:

Network Type	Required Action
Dynamic (IP addresses are assigned automatically by the network's DHCP (dynamic host connection protocol) server)	Make sure that the <i>Enabled DHCP</i> tick box is ticked (enabled).

Network Type	Required Action
Static (IP addresses are assigned manually)	Make sure that the <i>Enabled DHCP</i> tick box is un-ticked (disabled) and configure the settings described below: • IP address - Set the printer's IP address. Get an IP address from the network administrator and verify that the IP address is not used by another other device on the network. Note The first 3 sets of digits in the IP address must match other devices on the network that the printer is connected to. The fourth set of digits must be unique. For example, if the device that will transmit EDC data to the printer is connected to the same network and has the IP address 192.168.1.5, set the printer's IP address to 192.168.1.55 (if that address is not already in use). • Subnet mask - Set the subnet mask. The subnet mask must match other devices on the same network. • Default gateway - If the printer needs to communicate with devices on a different network (or the internet), enter the IP address of the network's default gateway (router).

3. If any settings were changed, select *Home > Setup > Production line setup > Power options > Restart* and wait for the printer to restart.

To setup and enable the Codenet connection via Ethernet:

1. Select *Home > Setup > Printer network > Protocol settings*.
2. Select the *Protocol Setting* drop down setting and select *Codenet*.
3. Un-tick (disable) the *Protocol enabled* tick box.
4. Select the *Protocol mode* drop down setting and select *TCP*.

Note The settings cannot be changed, when the protocol is enabled.

5. Settings described in the table below can now be defined. After these settings have been defined, tick (enable) the *Protocol Enabled* tick box.

Setting Name	Description
TCP Port	Define the port that will be used for codenet communication. Four TCP ports are available for Ethernet TCP/IP communication: 7000, 7001, 7002 and 7004 Default setting: 7000

Setting Name	Description
Response length	Define the length of the response data package: • Variable - ACK is one byte (0616), and NAK is four bytes (15, A, B, C). Where A, B and C represent a three character Hex error code. • Fixed - Both ACK and NAK are four bytes (06303030 and 15 A B C). Where A, B and C represent a three character Hex error code. Default setting: Variable
Send response	Define when a response to a data packet will be sent from the printer. • On Processed - The ACK or NAK will be sent when the printer has processed a data packet. • On Received - An ACK will be sent as soon as the printer receives a data packet. Note, the data has not yet been validated as correct. Default setting: On processed
LED state	This setting lets the LEDs on the printer cabinet to mimic the behavior of different printer types. This is useful if a different printer type is replaced with an Ax-Series printer. • Standard - The LEDs on the printer cabinet will mimic the behavior of A-Series i-Tech and A-Series plus printers. • Green + Flash - The LEDs on the printer cabinet will mimic the behavior of an A-Series classic printer. Default setting: Standard

Setting Name	Description
Printer ID	This setting allows the printer to respond using the ID of older printer types. This is useful if an older printer type is replaced with an Ax-Series printer, as the controller may be expecting a different response value. • 00 = Codebox • 01 = Solo • 02 = Solo Twin Head • 03 = A300 • 04-19 = Not Allocated • 20 = Macrojet • 21 = Casecoder • 22 = A-Series plus A100 • 23 = A-Series plus A300 • 24 = A-Series plus Duo • 25-29 = Not Allocated • 30 = Ax-Series • 31 - 99 = Not Allocated
Label format	Select which label format/raster will be used when a data packet containing a label design is sent to the printer.

Setting Name	Description
Backward compatible	This tick box changes the printer behavior if the 'R' (Clear all labels) command is used in between changing labels during continuous printing. Important: If the gap between prints is important, it is recommended to not use the 'R' (Clear all labels) command in between changing labels during continuous printing. Using this command introduces the risk that the gap between the last print of the old label and the first print of the new label will change. Backward Compatible Un-ticked (Disabled) - A print trigger is needed to re-start printing after the new continuous print label is put online: (1) Printer is in the "Ready" state. (2) Select and put online a label that has the "Repeat mode" set to "Continuous", send the command: 'P' (Put label with 3 digit name from store online) or 'ON' (Put label with variable length name from store online) (3) Send command: 'N' (Software Print Go) Note Printing starts. (4) Send command: 'R' (Clear all labels) Note A blank "NONE Continuous Printing Label" is put online which causes printing to stop. (5) Select and put online a different label that has the "Repeat mode" set to "Continuous", send the command: 'P' (Put label with 3 digit name from store online) or 'ON' (Put label with variable length name from store online) (6) The new label is not printed until the 'N' -(Software Print Go) command is sent again. Backward Compatible Ticked (Enabled) - Printing will start as soon as the new continuous print label is put online: (1) Printer is in the "Ready" state. (2) Select and put online a label that has the "Repeat mode" set to "Continuous", send the command: 'P' (Put label with 3 digit name from store online) or 'ON' (Put label with variable length name from store online) (3) To start printing, send the command: 'N' (Software Print Go) (4) Send command: 'R' (Clear all labels) Note A blank "NONE Continuous Printing Label" is put online which causes printing to stop. (5) Select and put online a different label that has the "Repeat mode" set to "Continuous", send the command: 'P' (Put label with 3 digit name from store online) or 'ON' (Put label with variable length name from store online)

Setting Name	Description
Backward compatible (Continued)	(6) Printing starts as soon as the new label is put online. The command 'N' (Software Print Go) is not needed to start printing. Default setting: Variable
Counter persistence on	Note This setting was added in software version 01.40.0629 When this tick box is selected, the values of counters in any labels newly created through Codenet will not be reset to the start value each time the label is sent to print. For example: (1) The counter value in the label reaches 30 and the label is taken offline. (2) The label is sent to print again, the counter value will advance to 31. Default setting: Ticked (Enabled)

6. Tick (enable) the *Protocol enabled* tick box.
7. To save the new settings, select *Home > Setup > Production line setup > Power options > Restart* and wait for the printer to restart.

CODENET COMMANDS

This next section of the manual describes the Codenet commands that are useful with EDC.

Codenet Command 'P'

This command is used to load a label to print, if the label name is a 3 digit number from 001 to 255.

When the command is received by the printer, the printer will respond with either a negative or positive acknowledgment.

Set format:

Start Transmission	Command ID	Head Select	Label Slot ID	End Transmission
ESC	P	1	001 to 255	EOT

Response format:

ACK = Positive acknowledgment, the command is received and implemented.

NAK = Negative acknowledgment, the command is received but failed.

Example:

Sent to printer: ESC P 1 001 EOT

Response from printer: ACK

Codenet Command 'ON'

This command is used to get a label with a variable length name and put it on-line for printing.

Set format:

Start Transmission	Command ID	Head Select	Length of Label Name	Label Name	End Transmission
ESC	ON	1	01 to 50	Any legal characters (Up to Length of Label Name)	EOT

Response format:

ACK = Positive acknowledgment, the command is received and implemented.

NAK = Negative acknowledgment, the command is received but failed.

Example:

Sent to printer: ESC ON 1 05 BEANS EOT

Response from printer: ACK

Codenet Command 'OE'

This command can be used to update data in the buffer queue that is used to populate EDC label elements. This command can also clear data from the buffer and history queue.

- Notes
1. It can be good practice to clear the buffer before sending data. This will make sure that no data in the buffer from a previous job is accidentally printed. However, only do this, if both the *Persistence enabled* and the *Buffer overwrite* settings are un-ticked (Disabled). Clearing the buffer excessively or unnecessarily can cause performance issues. If *Persistence enabled* is ticked (enabled), new data sent to the buffer will overwrite old data. Or, if *Persistence enabled* is un-ticked (disabled) and *Buffer overwrite* is ticked (Enabled), new data sent to the single spot in the buffer will overwrite old data.
 2. This command only works with ASCII data. To use Unicode, the data must be streamed into the printer's Ethernet socket.
 3. Each execution adds only one item to the queue at a time.
 4. If the queue is full, a NAK (negative acknowledgment) will be returned.

Set format:

Start Transmission	Command ID	Data Block Length	Data	End Transmission
ESC	OE	0001 to 1024	Any legal character (Up to Data Block Length)	EOT

Response format:

ACK = Positive acknowledgment, the command is received and implemented.

NAK = Negative acknowledgment, the command is received but failed.

Set Example:

Sent to printer: **ESC OE 0004 1234 EOT**

Response from printer: **06**

Clear buffer format:

Note The "Data Buffer Selection" part of the data string sets which buffer will be cleared. For example, 0 = TCP buffer, 1 = RS232 Buffer, 2 = History queue.

Start Transmission	Command ID	Clear Command	Data Buffer Selection	End Transmission
ESC	OE	0000	0, 1 or 2	EOT

Clear Example:

Sent to printer: **ESC OE 0000 1 EOT**

Response from printer: **06**

Domino Ax-Series Protocol Manual

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Domino UK Limited

Trafalgar Way
Bar Hill
Cambridge CB23 8TU
United Kingdom
Tel: +44 (0)1954 782551
Fax: +44 (0)1954 782874
Email: enquiries@domino-uk.com

Domino Amjet Inc.

1290 Lakeside Drive
Gurnee IL.60031
U.S.A.
Tel: +1 847 244 2501
Fax: +1 847 244 1421
Email: solutions@domino-na.com